

Steam Game Recommendation Using Machine Learning: A Comparative Analysis of Multi-Signal and Cold-Start Models

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Abstract—The expansive Steam catalogue presents significant information overload for users, complicated by cold-start challenges for new items and users. This paper presents a multi-signal recommendation system evaluated on the public Kozyriev Steam dataset (41.15M interactions). We integrate playtime, explicit recommendation votes, review helpfulness, and metadata tags into an implicit matrix factorization and learning-to-rank framework. Evaluating thirteen models under a time-aware full-catalogue top- N protocol ($K = 10$), Implicit Alternating Least Squares (ALS) and a tag-to-factor mapping (M3) jointly achieved the strongest ranking performance ($\text{Recall@10} \approx 0.0400$, $\text{NDCG@10} \approx 0.1218$), significantly outperforming Truncated SVD and popularity baselines ($p < 0.01$); ALS was selected as the deployed ranker for its native cold-user fold-in support. Multiplicative confidence weighting combining playtime and review helpfulness yielded measurable gains over binary labels. Furthermore, while global top-10 lists fail to surface cold items ($\text{HitRate@10} \approx 0$), dedicated surface partitioning enables effective cold-item retrieval ($\text{Recall@10} \approx 0.042$) and personalized new-release discovery. User cold-start evaluation demonstrates that personal preference fold-in stably surpasses popularity baselines after five seed likes.

Index Terms—Recommender systems, implicit feedback, matrix factorization, cold-start, learning-to-rank, Steam, collaborative filtering.

I. INTRODUCTION

A. Background and Motivation

With tens of thousands of video games available on Steam, manual navigation is inefficient. Recommender systems leverage historical interaction patterns via collaborative filtering (CF) to predict user preferences. However, standard CF relies strictly on interaction density, performing poorly when encountering “cold-start” items or users with sparse history. Existing Steam recommendation studies typically rely solely on binary play/ownership signals or friend networks. Playtime alone is noisy (e.g., idle client hours), while social graph data is often unavailable in public datasets.

Our work utilizes the Kaggle Kozyriev dataset, leveraging explicit recommend/not-recommend votes, review helpfulness ratings, playtime, and community metadata tags. By constructing a multi-signal continuous confidence weight and integrating tag-based content vectors, we model user preferences across both warm and cold interaction regimes.

B. Problem Formulation and Evaluation Design

We formulate game recommendation as a full-catalogue candidate ranking problem. Given a user’s historical positive interactions prior to a calendar cutoff, models generate a ranked list of all unobserved catalogue items. Because 85.78% of dataset reviews are positive (“recommended”), we evaluate models using top- N ranking metrics (Recall@K , NDCG@K , Catalogue Coverage, and Average Popularity Percentile) rather than accuracy or RMSE. To prevent temporal data leakage caused by non-stationary review patterns over time, we enforce a strict 80th-percentile temporal split date (2021-11-02), training strictly on historical data and evaluating on subsequent interactions.

II. RELATED WORK

Recent literature on Steam and digital game recommendation focuses on three primary paradigms:

- 1) *Collaborative and Neural Models*: Cheuque et al. [3] demonstrated that deep factorization models beat standard baselines on Steam data, though unstructured review text provided marginal lift. Bunga et al. [2] converted playtime into discrete star ratings for CF models (SlopeOne/SVD++), though discretization discards fine-grained interaction variance. Batra et al. [1] combined playtime and recommendation labels in Apache Spark ALS, but evaluated performance strictly via RMSE rather than top- N ranking.
- 2) *Graph Neural Networks and Cold Start*: Yang et al. [10] developed SCGRec, a social-aware contextual GNN

utilizing friend networks. Yang et al. [9] proposed GNN pre-training for item cold-start. Wang et al. [7] utilized deep autoencoders for profile-based recommendation. When social graphs are absent, alternative content representations are required.

- 3) *Hybrid Frameworks*: Gong et al. [4] and Wang et al. [8] developed hybrid CF-content models, emphasizing anti-popularity bias. Salkanović et al. [6] highlighted the trade-off between recommendation accuracy and catalogue coverage.

III. DATASET AND EXPLORATORY ANALYSIS

The Kozyriev Steam dataset [5] comprises 41,154,794 review records across 14.3M users and 50.8K games (Table I).

TABLE I
DATASET SUMMARY STATISTICS

Table	Rows	Columns
games.csv	50,872	13
users.csv	14,306,064	3
recommendations.csv	41,154,794	8
games_metadata.json	50,872	5 (incl. tags)

Due to memory constraints, dataset aggregation was performed via chunked streaming over recommendations.csv. Exploratory data analysis (EDA) revealed key structural characteristics:

- 1) *Temporal Non-Stationarity*: Review volume exhibits extreme growth over time (Fig. 1). A random train/test split would cause future review leakage into training matrices.
- 2) *Skewed Preference Distributions*: Playtime hours (Fig. 2–3) and review helpfulness votes (Fig. 4) exhibit severe right-skewness with heavy zero inflation. This motivates logarithmic transformation for preference weighting.
- 3) *Metadata Distribution*: Community game tags (Fig. 5) exhibit high concentration in popular action/indie categories, providing dense semantic representations for TF-IDF content vectors.

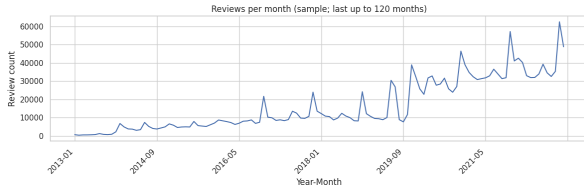


Fig. 1. Monthly review volume timeline (120-month subset), demonstrating strong temporal non-stationarity that necessitates time-aware train/test splitting.

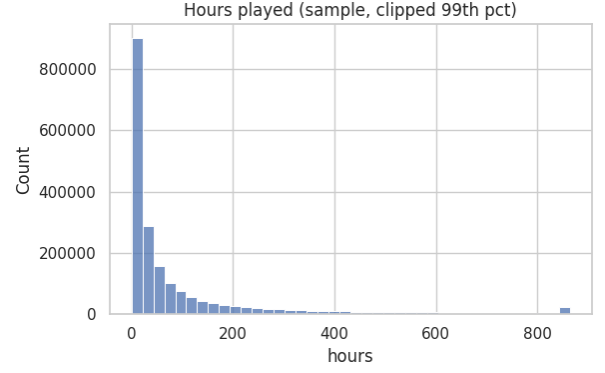


Fig. 2. Distribution of raw hours played (clipped at 99th percentile), exhibiting extreme right-skewness.

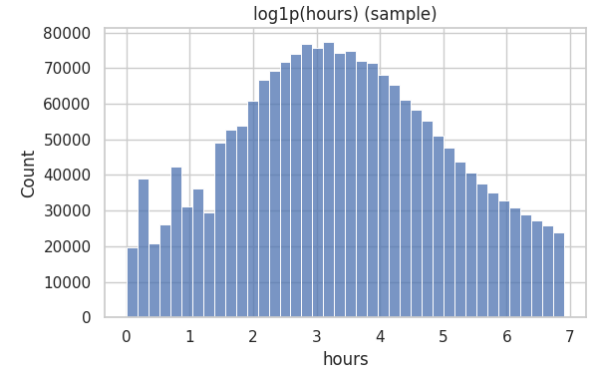


Fig. 3. Distribution of log-transformed playtime $\log(1 + \text{hours})$, illustrating log scaling for preference confidence weighting.

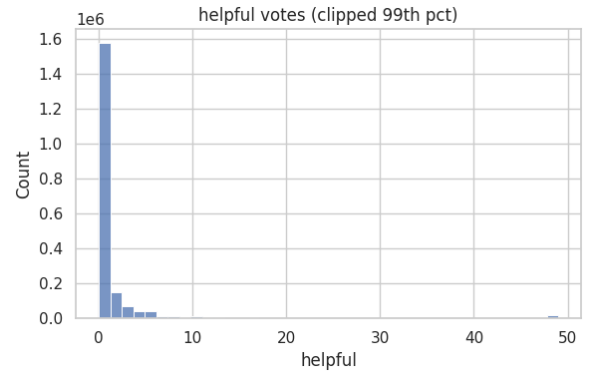


Fig. 4. Distribution of review helpfulness votes (clipped at 99th percentile), justifying continuous log-confidence weighting.

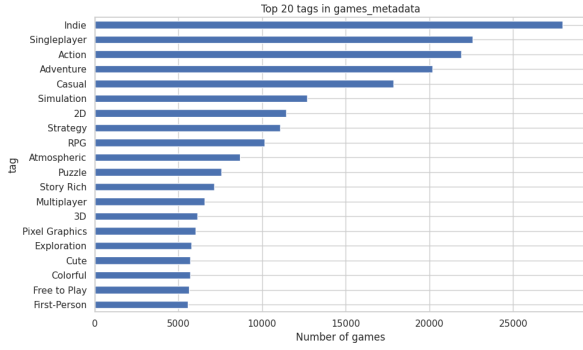


Fig. 5. Top 20 community game tags across catalogue metadata.

IV. METHODOLOGY

A. Preprocessing and Multi-Signal Confidence

Preprocessing executed in two full streaming passes over recommendations.csv. Pass 1 counted interactions per entity. Filtering thresholds were set at MIN_USER_REVIEWS = 10 and MIN_GAME_REVIEWS = 30, capping at 80,000 users and 12,000 games. Pass 2 materialized 4,830,132 dense interaction rows, downsampled to 4,000,006 rows across 56,291 users to preserve complete user interaction histories.

A calendar split at 2021-11-02 (80th percentile) yielded 3,200,557 training interactions (55,898 users, 10,738 games) and 799,449 test interactions. For positive interactions (is_recommended = True), continuous confidence weights were constructed as:

$$c_{u,i} = \log(1 + \text{hours}_{u,i}) \times \log(1 + (1 + \text{helpful}_{u,i})) \quad (1)$$

Rows with is_recommended = False (593,014 training rows, 18.5%) were preserved as explicit negative evidence for targeted experiments. A TF-IDF tag matrix was constructed across 300 top metadata tags for 10,738 games.

B. Model Taxonomy

Thirteen models across baselines, matrix factorization, learning-to-rank (LTR), and hybrid approaches were evaluated (Table II).

TABLE II
EVALUATED MODEL ARCHITECTURE CHEAT-SHEET

Code	Architecture / Algorithm Description
B1	Popularity ranker (training positive count)
B2	Tag TF-IDF content profile cosine similarity
B3	Truncated SVD (64 components) on confidence matrix
B4	Hand-weighted hybrid blend (Popularity, Content, SVD)
B5	Hybrid blend with reserved cold-item allocation slots
M1	Implicit ALS ($\alpha = 20.0$, $\lambda = 0.05$, 64 latent factors)
M2a	LightGBM LambdaRank (tag-retrieval negative sampling)
M2b	LightGBM LambdaRank (uniform + $\text{pop}^{0.75}$ negatives)
M2c	M2b + explicit negative profile features & hard negatives
M2d	M2c + release recency features & age flags
M3	Ridge mapping from TF-IDF tags to M1 ALS factor space
M4	M1 ALS with tuned explicit dislike penalty ($\lambda \approx 0$)
Two-Stage	Tag retrieval shortlist re-ranked via M2b LTR

1) *Implicit ALS (M1)*: The core collaborative ranker minimizes the implicit matrix factorization loss:

$$\min_{x_u, y_i} \sum_{u,i} c_{u,i} (p_{u,i} - x_u^T y_i)^2 + \lambda \left(\sum_u \|x_u\|^2 + \sum_i \|y_i\|^2 \right) \quad (2)$$

where $p_{u,i} = 1$ for positive pairs, producing latent factor matrices $X \in \mathbb{R}^{55,898 \times 64}$ and $Y \in \mathbb{R}^{10,738 \times 64}$.

2) *Cold-User Fold-In*: To serve new onboarding users with I_u seed likes without retraining, exact iALS solver fold-in was computed over frozen item factors Y :

$$x_u = \left(Y^T Y + \sum_{i \in I_u} (c_i - 1) y_i y_i^T + \lambda I \right)^{-1} \sum_{i \in I_u} c_i y_i \quad (3)$$

V. RESULTS AND EVALUATION

A. Evaluation Protocol and Metrics

Evaluation was conducted on a fixed cohort of 572 held-out users (300 warm, 272 cold) scoring all 10,738 unseen training games. Metrics include Recall@K, NDCG@K, HitRate@K, Catalogue Coverage, and Average Popularity Percentile. Paired statistical significance was computed using 10,000-resample bootstrap 95% confidence intervals, Wilcoxon signed-rank tests, and Holm-Bonferroni corrections.

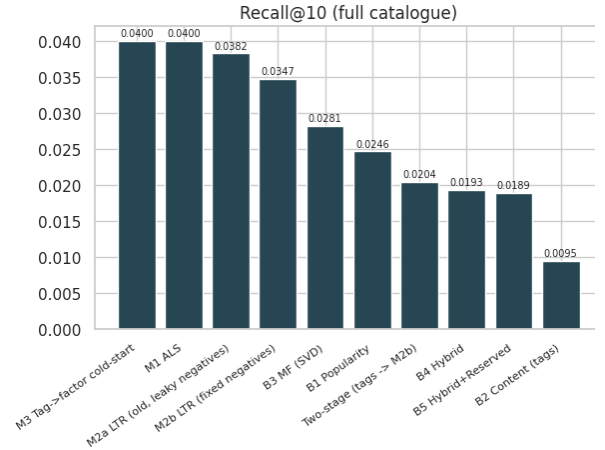


Fig. 6. Global catalogue Recall@10 across models (572 evaluation users). M1 ALS and M3 lead overall ranking performance.

B. Comparative Model Performance

Table III and Fig. 6–8 summarize full-catalogue top-10 metrics across all thirteen models. M1 ALS, M3 (tag-to-factor map), and M4 (ALS + dislike penalty) achieved statistically indistinguishable top ranking performance (Recall@10 $\approx$ 0.0400, NDCG@10 $\approx$ 0.1218; paired test M3 vs. M1, $\Delta = 0$, Holm $p = 1$). We ship M1 as the primary ranker despite this tie because it alone supports the closed-form fold-in used for cold-user onboarding (Sec. IV-B2), while M3's tag-to-factor mapping is retained specifically for the cold-item surface below.

Paired significance tests confirm M1 ALS significantly outranked B1 Popularity ($\Delta \text{Recall@10} = +0.0153$, $p < 10^{-4}$).

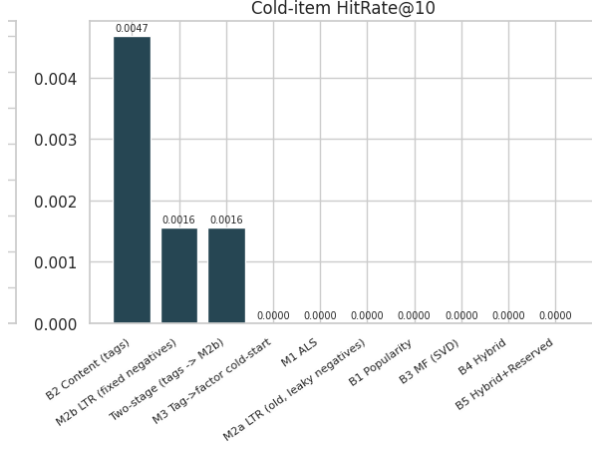


Fig. 7. Main-list Cold-item HitRate@10 across models, showing cold items rarely surface on the global top-10 list.

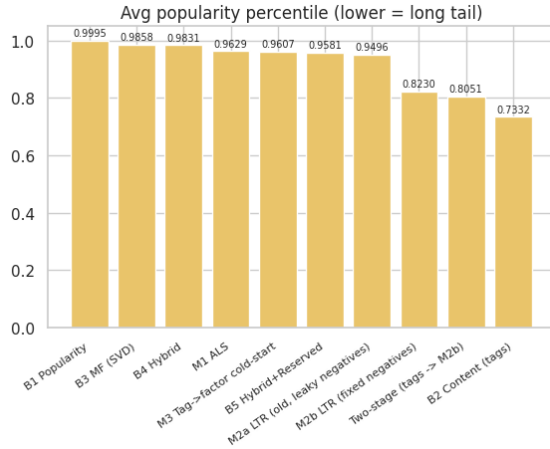


Fig. 8. Average popularity percentile across models. Personalized rankers significantly reduce blockbuster bias.

TABLE III
FULL-CATALOGUE TOP-10 BENCHMARK (572 USERS, $K = 10$)

Model	Recall@10	NDCG@10	HitRate@10	Coverage	Avg Pop %
B1 (Popularity)	0.0246	0.0767	0.3969	0.0022	99.95
B2 (Content)	0.0095	0.0326	0.1573	0.2018	73.32
B3 (Truncated SVD)	0.0281	0.0965	0.4196	0.0306	98.58
B4 (Hybrid Blend)	0.0193	0.0623	0.3199	0.0327	98.31
B5 (Cold Slot Blend)	0.0189	0.0610	0.3094	0.0350	95.81
M1 (Implicit ALS)	0.0400	0.1218	0.4685	0.1155	96.29
M2a (LTR Tag Neg)	0.0382	0.1073	0.4580	0.0965	94.96
M2b (LTR Matched)	0.0347	0.0920	0.4476	0.1450	82.30
M2c (LTR Neg Feat)	0.0323	0.0912	0.4371	0.1386	88.14
M2d (LTR Recency)	0.0326	0.0857	0.4266	0.1284	83.76
M3 (Tag-Factor Map)	0.0400	0.1218	0.4685	0.1160	96.07
M4 (Dislike Penalty)	0.0400	0.1216	0.4685	0.1148	96.35
Two-Stage	0.0204	0.0521	0.2570	0.1850	80.51

and B3 SVD ($\Delta\text{Recall@10} = +0.0119, p = 0.006$). Feature gain analysis on LightGBM LTR showed tag cosine (64.1%) and ALS score (28.2%) dominated LTR performance, confirming that complex LTR trees mostly re-encoded signals already present in M1/B2.

C. Multi-Surface and Cold-Start Evaluation

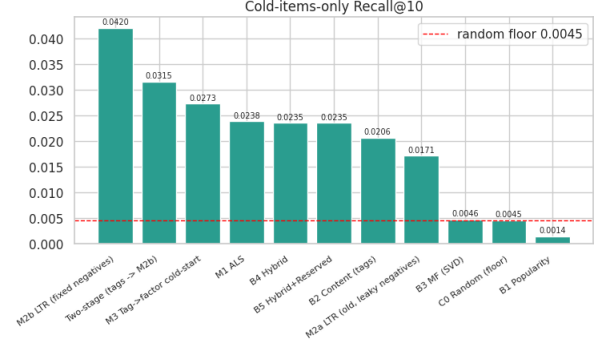


Fig. 9. Cold-items-only surface Recall@K across evaluation depths (2,761 cold items). M2b LTR leads (Recall@10 ≈ 0.042), 9 $\times$ above the random floor.

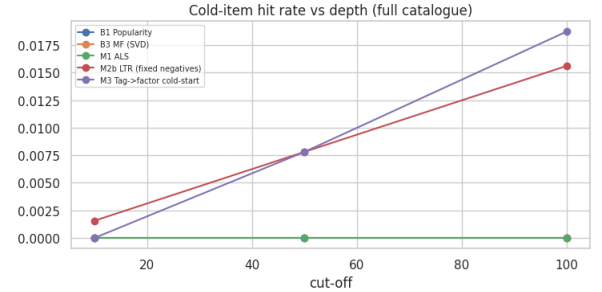


Fig. 10. Cold-item hit rate vs. evaluation depth on the full catalogue.

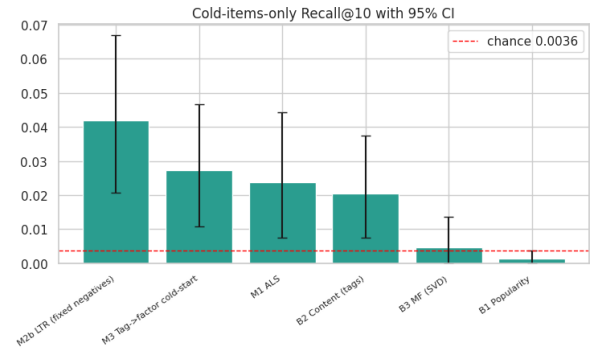


Fig. 11. Cluster-bootstrap 95% confidence intervals for cold-items-only Recall@10 across models.

1) *Cold-Item Surface*: Main catalogue cold-item HitRate@10 is ≈ 0 across all global rankers. However, evaluating on a dedicated cold-item surface (2,761 candidate items, Fig.

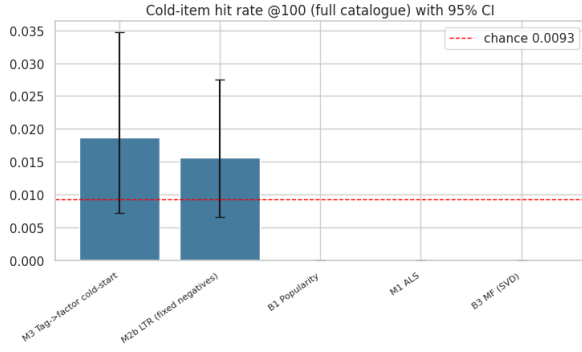


Fig. 12. Cluster-bootstrap 95% confidence intervals for cold-item hit rate @10 on the global catalogue.

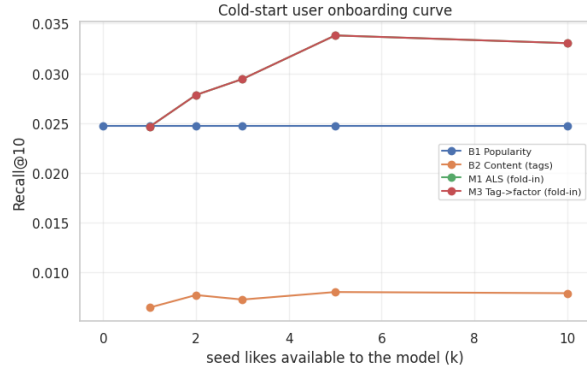


Fig. 13. Cold-user onboarding curve: Recall@10 vs. seed likes $k \in [0, 10]$. ALS fold-in surpasses popularity from $k \geq 5$.

9–12) achieved Recall@10 ≈ 0.042 via M2b LTR (9 $\times$ random floor 0.0045).

- 2) *New-Release Surface*: Personalized ALS achieved Recall@10 = 0.1603 on 1,588 new titles (< 365 days old), outperforming date-sorted ranking (Recall@10 = 0.0222). Note that only 762 of these 1,588 titles are also cold-start items, so this result chiefly reflects a recent-releases shelf rather than a zero-history cold-start test.
- 3) *Cold-User Onboarding*: User fold-in evaluation (566 simu-

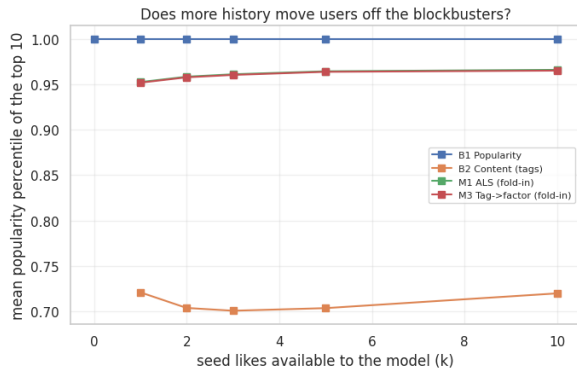


Fig. 14. Cold-user onboarding curve: Average popularity percentile vs. seed likes $k \in [0, 10]$.

lated new users, Fig. 13–14) showed ALS fold-in reliably overtakes popularity from $k \geq 5$ seed likes onward.

VI. DISCUSSION

A. Comparative Improvements Over Reference Papers

Our findings advance baseline Steam recommendation research in four key areas:

- 1) *Multi-Signal Confidence Weighting*: In controlled ablations, our multiplicative formula ($c_{u,i}$) achieved Recall@10 = 0.0400, outperforming binary ALS (0.0377), hours-only (0.0363), helpfulness-only (0.0376), and additive weighting (0.0348). This contrasts with discrete rating binning (Bunga et al. [2]) and unweighted CF.
- 2) *Leakage-Free Evaluation*: By enforcing a temporal cutoff (2021-11-02), we eliminate data leakage present in random splits (Batra et al. [1], Cheuque et al. [3]).
- 3) *Disentangled Cold-Start Surface Modeling*: We demonstrate that cold-start is not a monolithic problem: main-list cold items require separate surfaces, whereas cold users require seed-threshold onboarding ($k \geq 5$).
- 4) *Rigorous Statistical Protocol*: Paired 10,000-resample bootstrap CIs and Holm-Bonferroni corrections confirm true performance gaps across 13 models.

To demonstrate real-world deployment viability, we instantiated our trained implicit ALS and fold-in model into an interactive web application (built with Flask/Python and JavaScript) offering real-time user onboarding and four distinct recommendation surfaces.

B. Limitations

- 1) *Model Scope*: Deep neural models (DeepFM, GNNs) were not implemented due to dataset structure and lack of social graph data (unlike SCGRec [10]).
- 2) *Dense Subset Restriction*: Filtering users (≥ 10 reviews) and games (≥ 30 reviews) limits findings to active populations.
- 3) *Explicit Negative Feedback*: Incorporating 593K negative reviews (M4 penalty, M2c hard negatives) yielded no statistical ranking improvement over positive-only ALS.
- 4) *Tag Metadata Sparsity*: Games with empty tag lists cannot generate tag-overlap explanations.

C. Future Work

Future enhancements include: (1) extracting text embeddings from game descriptions using transformers (BERT/LLMs) to eliminate tag sparsity; (2) implementing GNN frameworks (LightGCN) on bipartite user-item-tag graphs; (3) incorporating exponential time-decay factors into interaction confidence; and (4) conducting paired bootstrap evaluation across larger cold-item user samples.

VII. CONCLUSION

This paper evaluated thirteen recommendation algorithms on a dense 4M-row temporal split of the Steam Kozyriev dataset. Multi-signal implicit ALS with multiplicative playtime-helpfulness confidence, tied with a tag-to-factor mapping (M3), achieved the strongest top-10 ranking accuracy among

all models ($\text{Recall@10} \approx 0.0400$, $\text{NDCG@10} \approx 0.1218$), significantly outperforming SVD and popularity baselines; ALS was deployed as the shipped ranker for its fold-in support. Surface disentanglement demonstrated that cold items require dedicated recommendation shelves, while cold-user fold-in successfully personalizes recommendations after five seed likes.

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